CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342-Section
SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: Mini Inventory Management System

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Version: 2

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1. Introduction

1.1. Purpose

The Mini Inventory Management System is designed to help small to medium businesses manage their inventory more efficiently by digitizing stock tracking. It reduces errors caused by manual paper or spreadsheet-based tracking, provides real-time visibility of stock levels, and supports better decision-making for purchasing and restocking. The system is intended for small business owners, store managers, or office administrators who need an effective way to control inventory without relying on complex enterprise systems.

1.2. Scope

The Mini Inventory Management System is designed to manage the inventory of products for small to medium businesses, providing a digital alternative to manual record-keeping. The system allows staff to perform stock in and stock out operations through an Android application, while administrators can manage products, categories, and user accounts through a web-based dashboard. It maintains a history of all inventory transactions, enabling accountability and real-time tracking of stock levels.

Boundaries of the system include:

- The system does not handle sales transactions, supplier management, or financial accounting.
- It requires an internet connection for synchronization between the Android app and the web system.
- Only authorized users (staff and admins) can access the system features.
- Reports generated are limited to inventory summaries and stock logs; advanced analytics or forecasting are not included.

The system focuses on accuracy, efficiency, and accountability in inventory management while providing a simple and scalable solution for small businesses.

1.3. Definitions, Acronyms, and Abbreviations

Term	Definition
Admin	Administrator user with full system access and management capabilities
Staff	Regular user responsible for stock tracking and inventory updates
Stock In	Transaction to add inventory/products to the system
Stock Out	Transaction to remove inventory/products from the system
Product	Individual item managed by the inventory system
Category	Classification or grouping of similar products
Stock Log	Record of all inventory transactions (additions and removals)
Dashboard	Web-based interface for administrators to manage the system

Android App	Mobile application used by staff for inventory updates
Real-time Synchronization	Immediate update of data across all system modules
SRS	Software Requirements Specification
API	Application Programming Interface for communication between app and server
Database	Centralized data storage containing products, categories, users, and logs
Authentication	Process of verifying user identity through login credentials
Authorization	Process of determining what authenticated users are allowed to do

2. Overall Description

2.1. System Perspective

The Mini Inventory Management System is a standalone solution that helps small to medium businesses manage inventory digitally. Staff use an Android app to update stock, while admins use a web dashboard to manage products, categories, and users. Both modules connect to a centralized database, ensuring real-time synchronization, accuracy, and accountability in inventory management.

2.2. User Classes and Characteristics

Admin

- Responsible for managing products, categories, and user accounts
- Full access to the web dashboard
- Can generate inventory reports
- Typically business owners or managers
- Requires moderate to advanced computer skills
- Performs periodic management and configuration tasks

Staff

- Responsible for updating stock levels through the Android app
- Can perform stock in/out actions and view current product quantities
- Cannot manage system settings or user accounts
- May have varying levels of technical proficiency
- Performs frequent daily inventory updates

2.3. Operating Environment

The system operates in a client-server environment with the following requirements:

Hardware:

- Android smartphone (for staff) – minimum 2GB RAM
- PC or laptop (for admin) – standard business computer
- Server – with sufficient processing power and storage capacity for the database

Software:

- Android OS 8.0 or higher for the mobile app
- Modern web browsers (Chrome, Firefox, Safari, Edge) for the web dashboard
- Backend server infrastructure with database management system
- Web server

Network:

- Stable internet connection for both app and web dashboard
- HTTPS protocol for secure data transmission

2.4. Assumptions and Dependencies

Assumptions:

- Users have basic knowledge of smartphones and computers
- Product information is entered accurately by staff to maintain correct inventory levels
- The organization has access to necessary hardware and internet connectivity
- User roles and responsibilities are clearly defined and understood
- Regular database backups are performed by the system administrator

Dependencies:

- Internet connection is required for real-time updates between the Android app and web dashboard
- The database server is accessible, secure, and properly maintained
- Compatible hardware and software are available for proper operation
- Third-party authentication or payment systems (if integrated in the future)
- Regular software updates and security patches are applied

2.4 Assumptions and Dependencies

- Users have basic knowledge of smartphones and computers.
- Internet connection is required for real-time updates between the Android app and web dashboard.
- The database server is accessible and secure.
- Product information is entered accurately by staff to maintain correct inventory levels.
- The system depends on compatible hardware and software for proper operation.

3. System Features and Functional Requirements

3.1. Feature 1: Stock In / Stock Out

Description: This feature allows staff to update product quantities by adding or removing stock through the Android app. Every transaction is recorded in the database for accountability and traceability.

Functional Requirements:

- FR1.1: Staff must log in with valid credentials to access stock functions
- FR1.2: Staff can select a product from an active product list
- FR1.3: Staff can enter the quantity to add (Stock In) or remove (Stock Out)
- FR1.4: The system validates that the quantity does not exceed available stock for Stock Out operations
- FR1.5: The system displays a confirmation message showing the transaction details before finalizing
- FR1.6: The system updates the product quantity in the Products table immediately upon confirmation
- FR1.7: The system creates a Stock_Log entry with timestamp, user ID, product ID, quantity changed, and transaction type
- FR1.8: Staff can view the current stock level of any product before performing a transaction
- FR1.9: The system prevents negative stock levels (cannot remove more than available)
- FR1.10: Transactions are synced with the web dashboard in real-time

3.2. Feature 2: Product and Category Management

Description: This feature allows administrators to manage products and categories through the web dashboard, including adding, editing, deleting, and organizing entries. Admins can also assign products to appropriate categories and manage product details.

Functional Requirements:

- FR2.1: Admin must log in with valid credentials to access product and category management
- FR2.2: Admin can create new categories with a name and description
- FR2.3: Admin can edit existing category information
- FR2.4: Admin can delete categories (with confirmation and validation that no products are assigned)
- FR2.5: Admin can create new products with details including name, SKU, description, and category assignment
- FR2.6: Admin can edit product information including name, category, description, and other details
- FR2.7: Admin can delete products (with proper audit trail in the system)
- FR2.8: Admin can assign or reassign products to categories
- FR2.9: All product changes are reflected in real-time on the Android app
- FR2.10: The system maintains an audit trail of who modified what and when

3.3. Feature 3: User Authentication and Access Control

Description: This feature manages user login, authentication, and role-based access control to ensure only authorized users can perform specific actions in the system.

Functional Requirements:

- FR3.1: Users must provide a username and password to access the system
- FR3.2: The system validates credentials against the user database
- FR3.3: Invalid login attempts are logged for security monitoring
- FR3.4: Users are locked out after multiple failed login attempts (e.g., 5 attempts)
- FR3.5: Passwords are encrypted and stored securely in the database
- FR3.6: The system enforces password requirements (minimum length, complexity)

- FR3.7: Users can change their password through their profile
- FR3.8: Admin can reset user passwords and manage user accounts
- FR3.9: The system assigns roles (Admin or Staff) to each user
- FR3.10: Access to features is enforced based on user role
- FR3.11: Session tokens expire after a period of inactivity
- FR3.12: Users are prompted to log out, and sessions are terminated securely

3.4. Feature 4: Inventory Reporting and Analytics

Description: This feature provides administrators with reports on inventory status, stock levels, transaction history, and system usage for better decision-making.

Functional Requirements:

- FR4.1: Admin can generate a product inventory summary showing current stock levels
- FR4.2: Admin can generate a stock transaction log showing all stock in/out operations
- FR4.3: Reports can be filtered by date range, product, category, or user
- FR4.4: Admin can export reports in PDF and CSV formats
- FR4.5: The system displays the date and time of report generation
- FR4.6: Admin can view transaction history with details including product, quantity, user, and timestamp
- FR4.7: The system calculates and displays product stock age or turnover rates
- FR4.8: Admin can view a list of low-stock items based on customizable thresholds
- FR4.9: Reports include user activity logs showing who performed each transaction
- FR4.10: The system archives reports for historical reference

3.5. Feature 5: Real-time Data Synchronization

Description: This feature ensures that data is synchronized in real-time between the Android app and the web dashboard, maintaining consistency and accuracy across all system components.

Functional Requirements:

- FR5.1: Changes made on the Android app are synchronized to the database within seconds
- FR5.2: Changes made on the web dashboard are reflected on the Android app in real-time
- FR5.3: The system handles offline scenarios gracefully (queues transactions for sync when online)
- FR5.4: The system detects and resolves data conflicts if simultaneous updates occur
- FR5.5: Sync status is visible to users (showing whether the app is synced or pending sync)
- FR5.6: The system logs all synchronization events for auditing purposes
- FR5.7: Users are notified if synchronization fails
- FR5.8: The system automatically retries failed synchronization attempts

4. Non-Functional Requirements

- Performance: The system should handle multiple users updating stock simultaneously without delays.
- Security: User passwords are encrypted, and access is restricted based on roles (Admin or Staff).
- Usability: Both the Android app and web dashboard should be intuitive and easy to navigate.
- Reliability: All stock transactions must be accurately recorded, and data must remain consistent.
- Scalability: The system should support additional products, users, and stock logs as the business grows.

- Maintainability: The code should follow standard conventions to allow easy updates and future enhancements

5. System Models (Diagrams)

5.1. ERD

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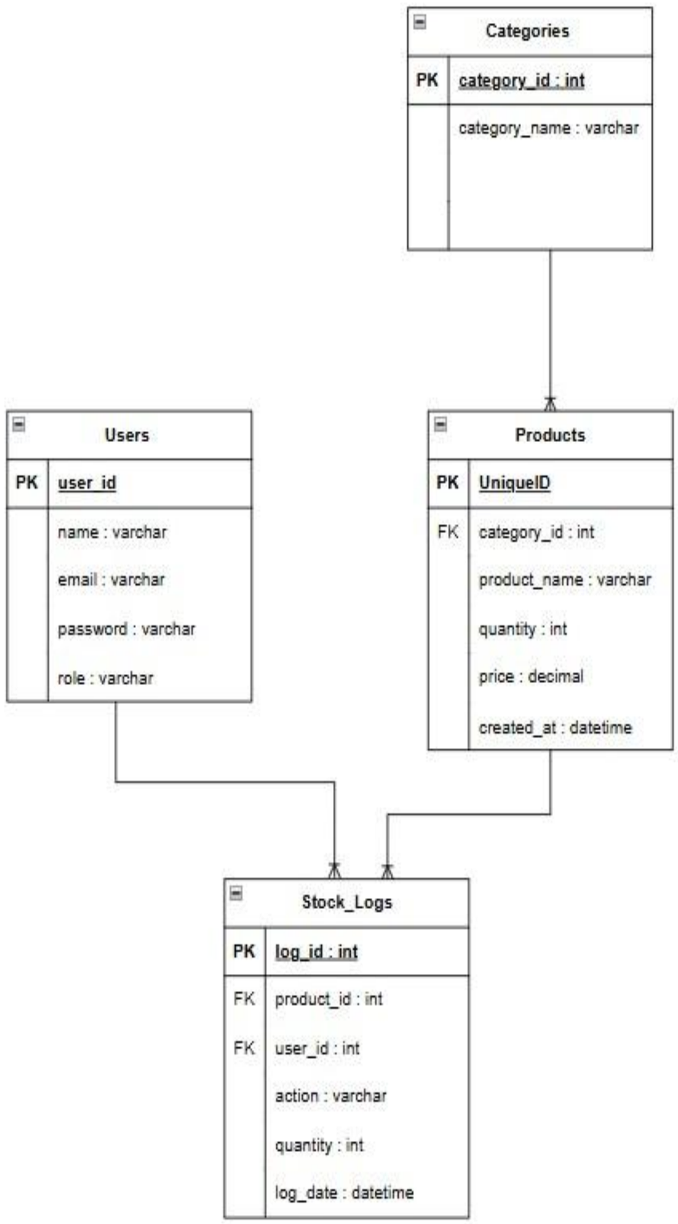
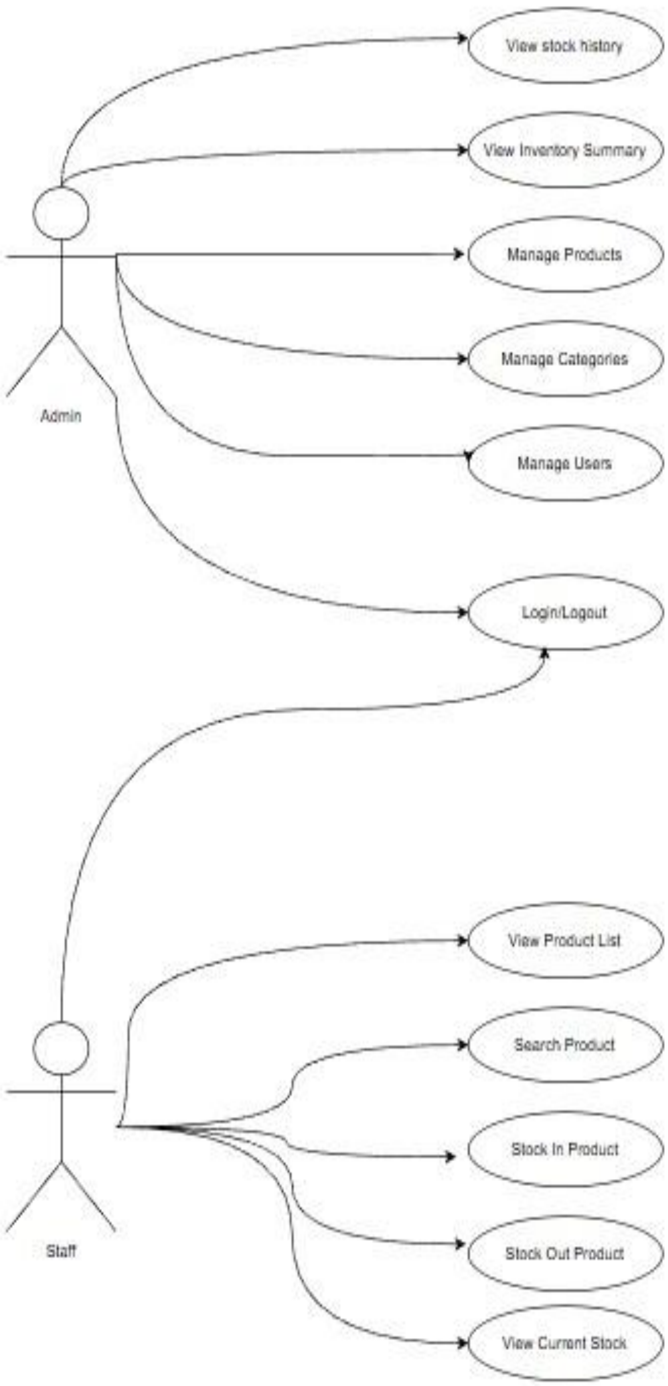
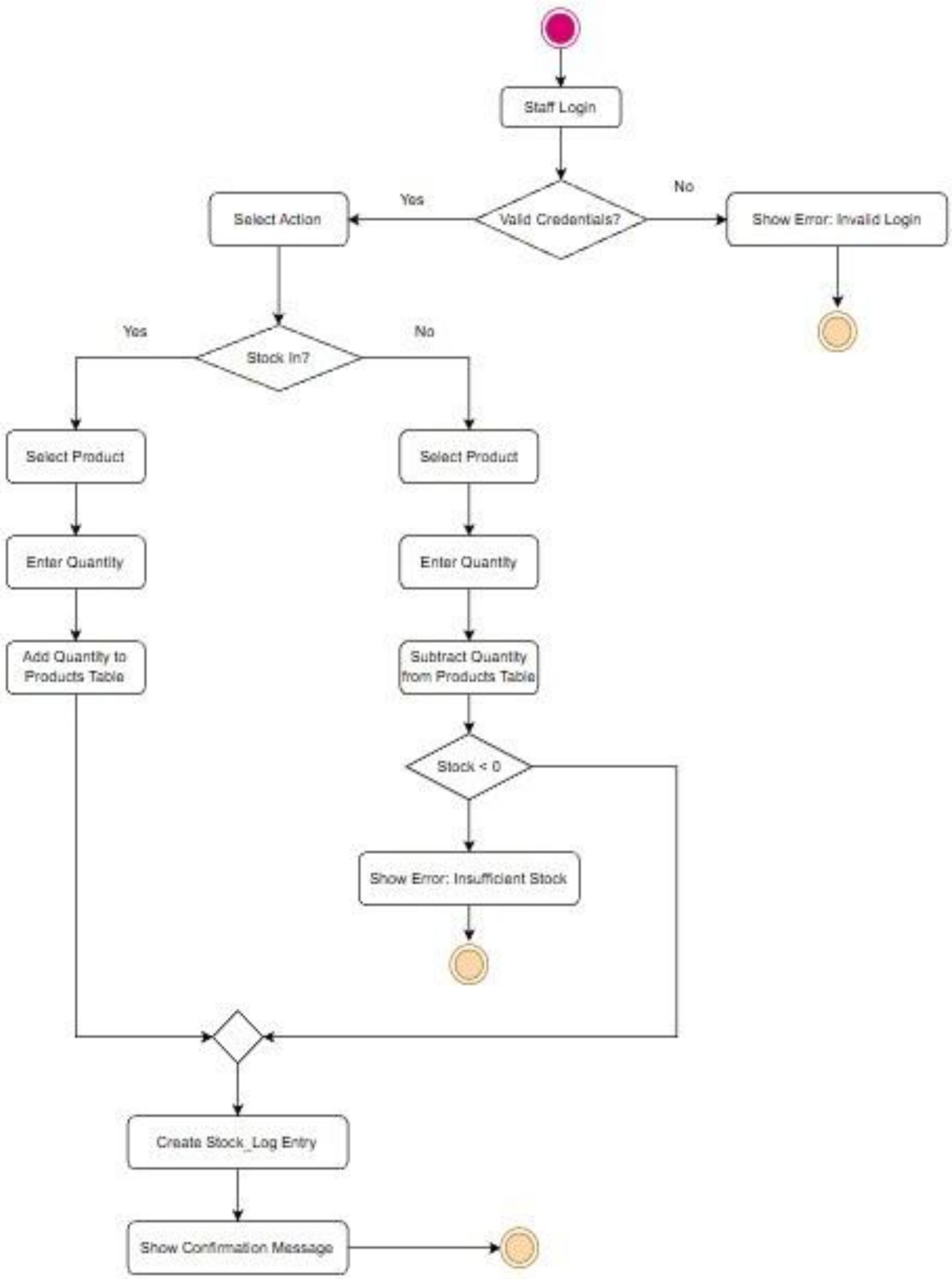


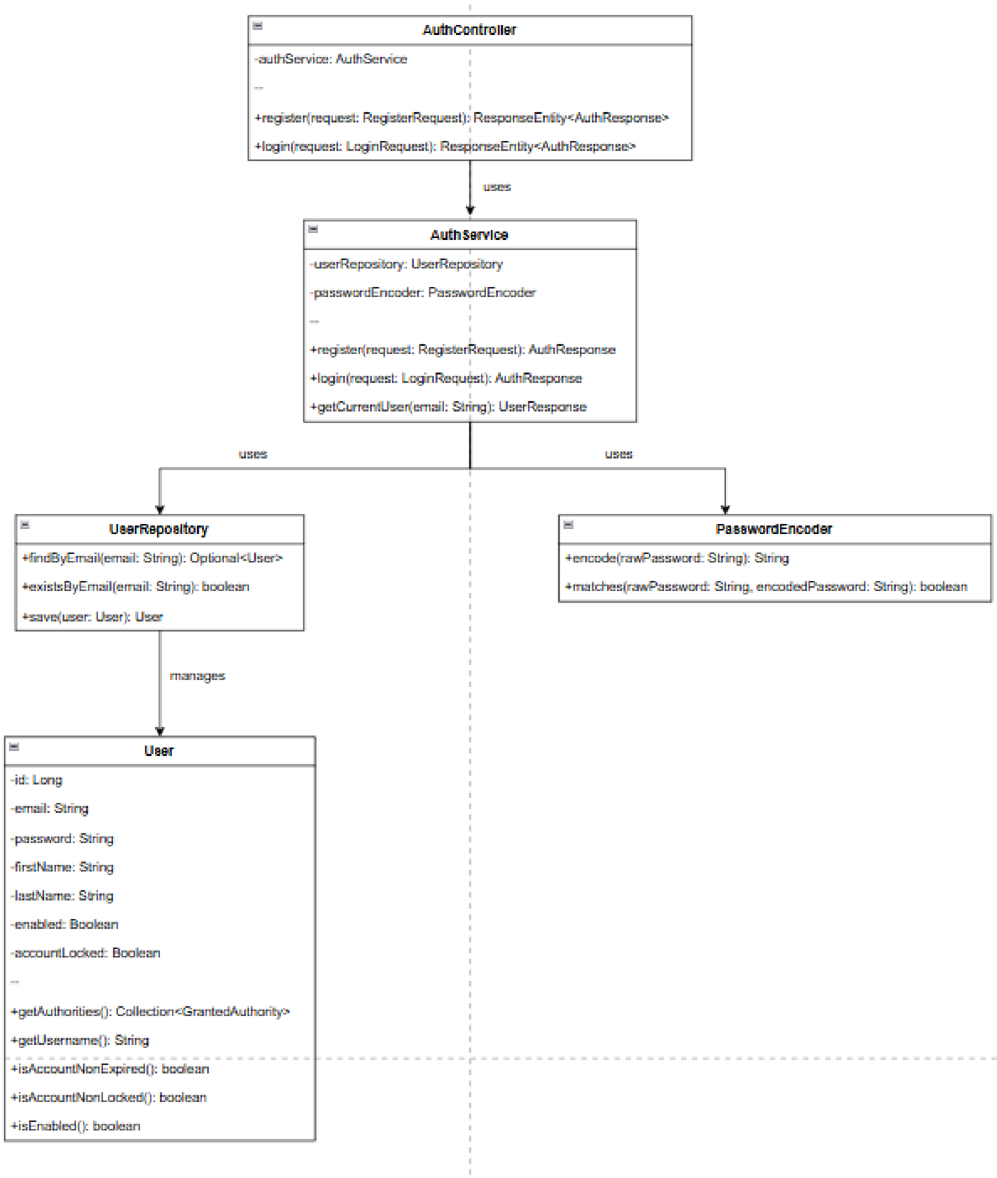
5.2. Use Case Diagram



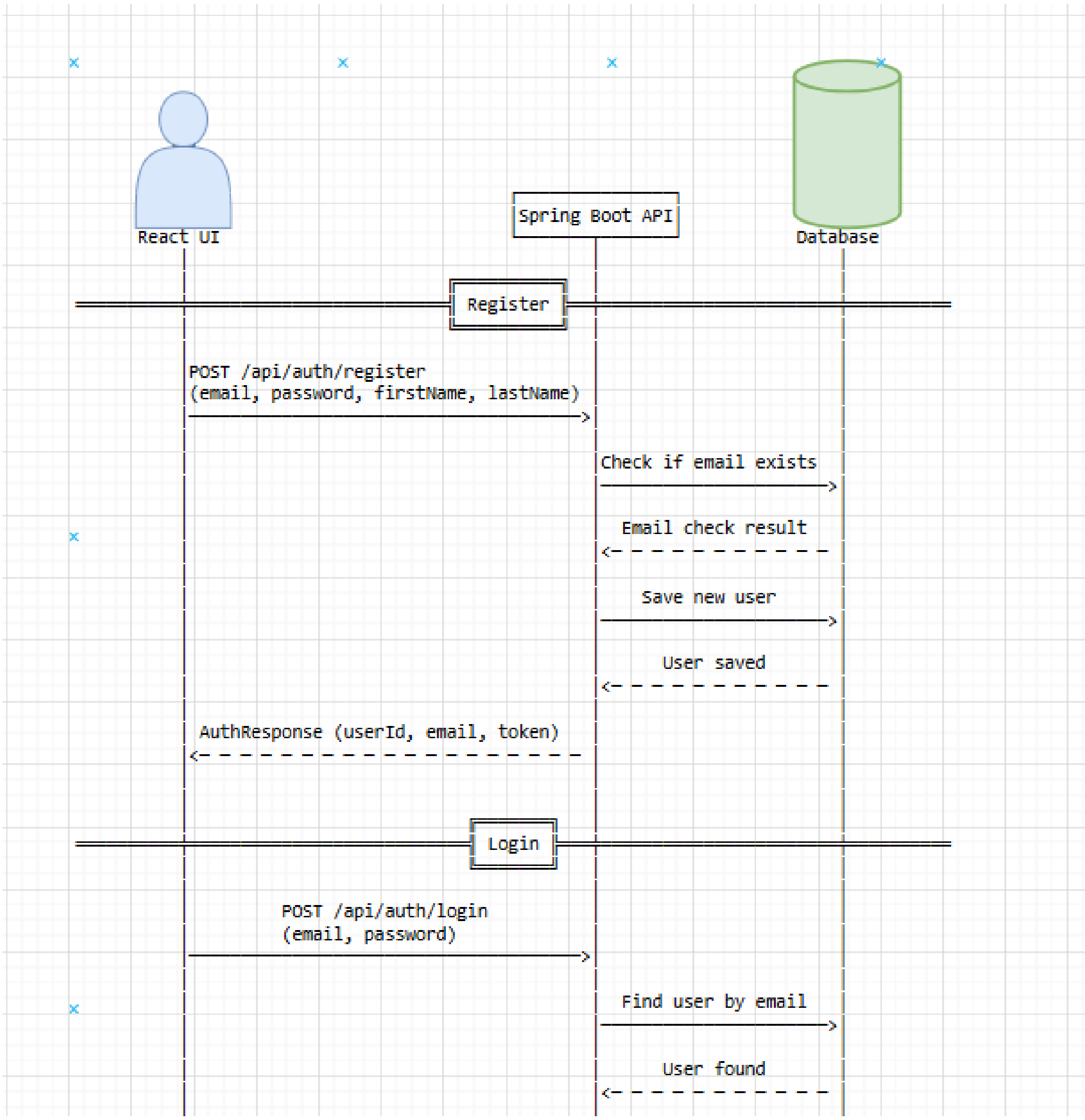
5.5. Activity Diagram

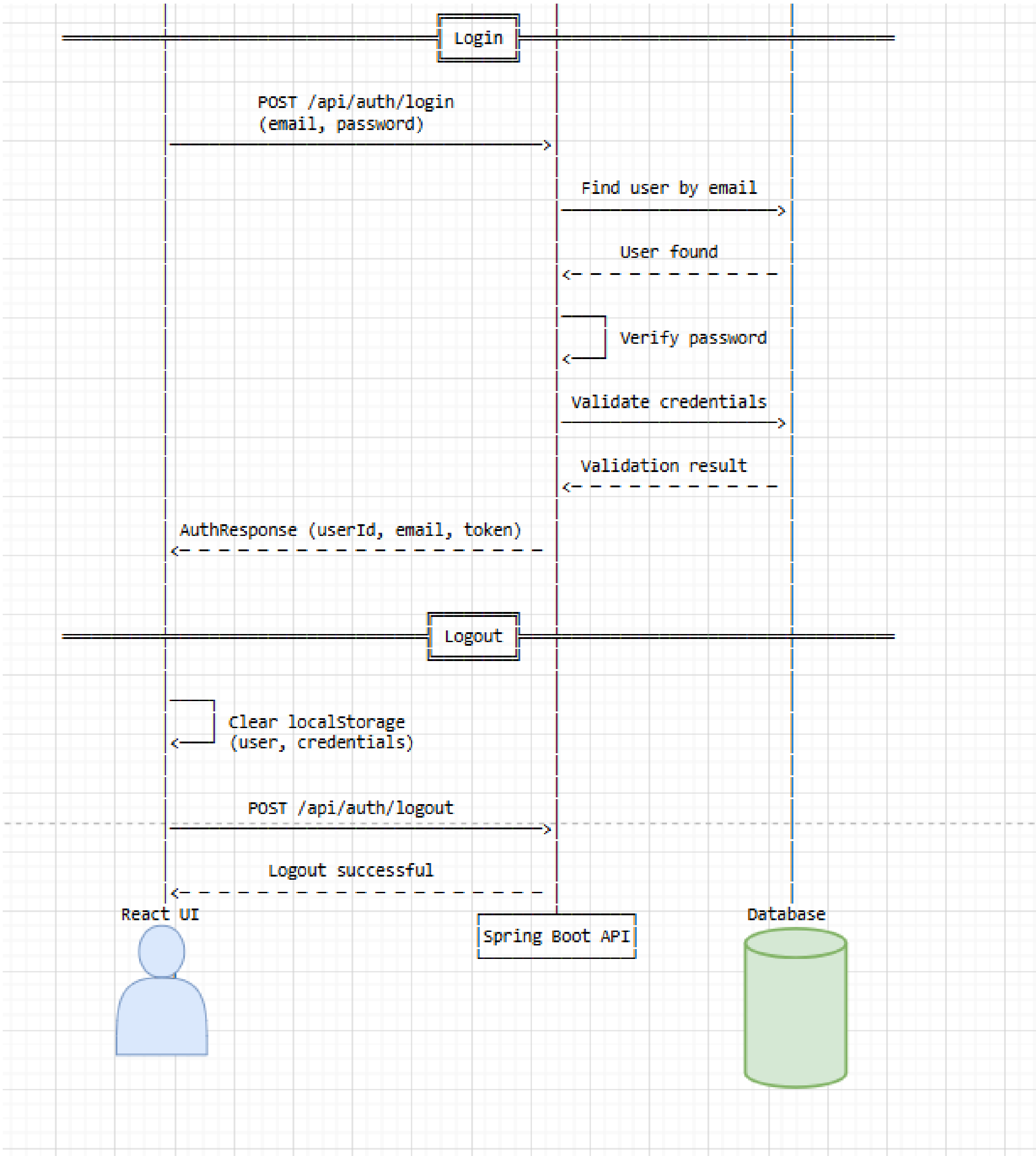


5.7. Class Diagram



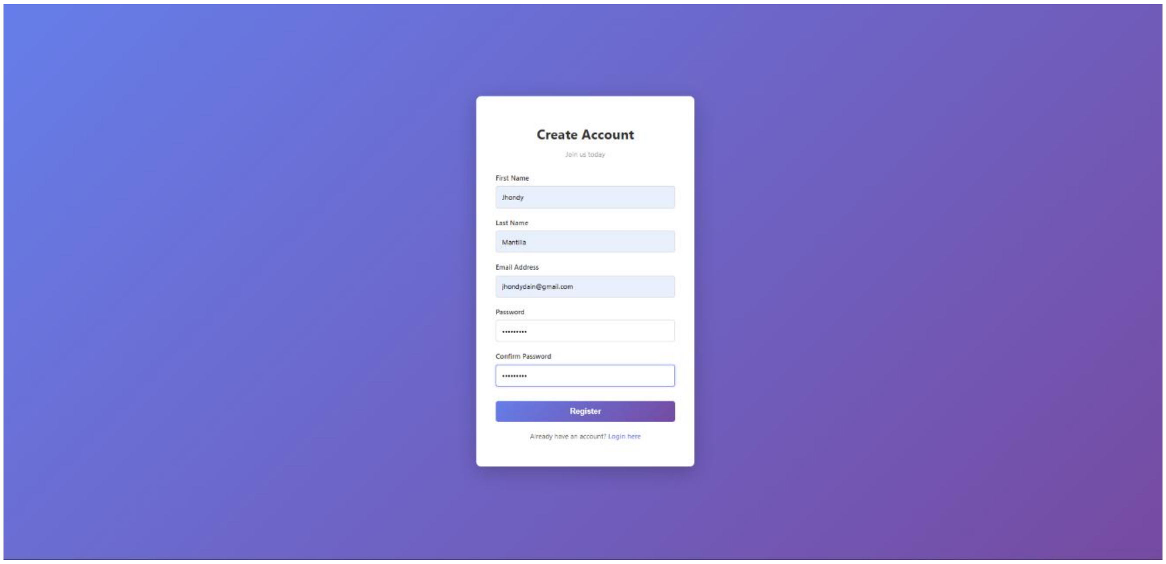
5.8. Sequence Diagram



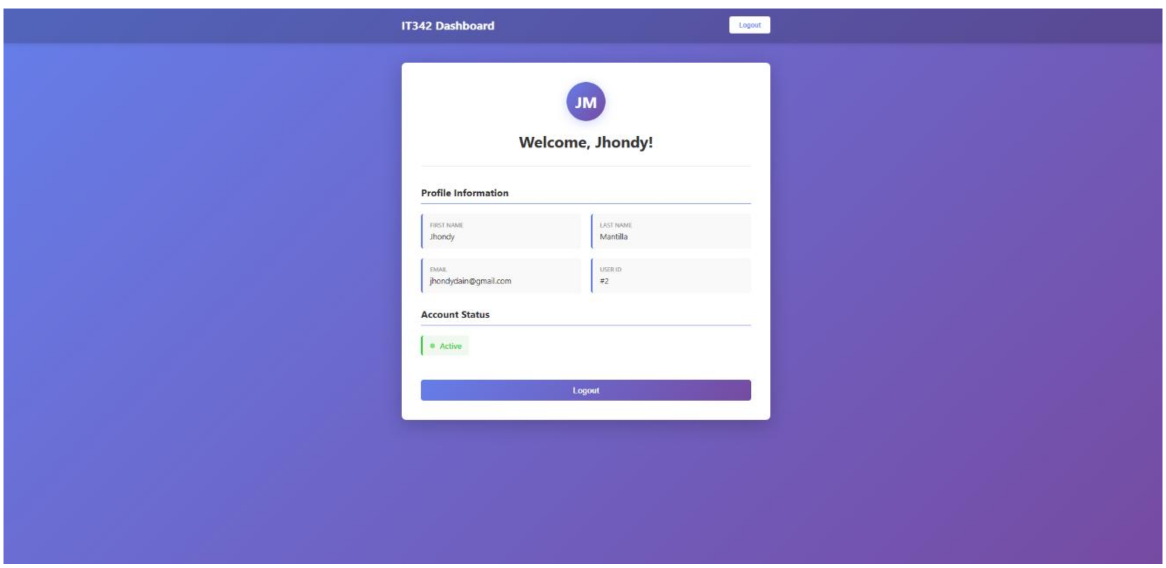

Login
POST /api/auth/login
(email, password)
Find user by email
User found
Verify password
Validate credentials
Validation result
AuthResponse (userId, email, token)
Logout
Clear localStorage
(user, credentials)
POST /api/auth/logout
Logout successful
React UI
Spring Boot API
Database

Web screenshots

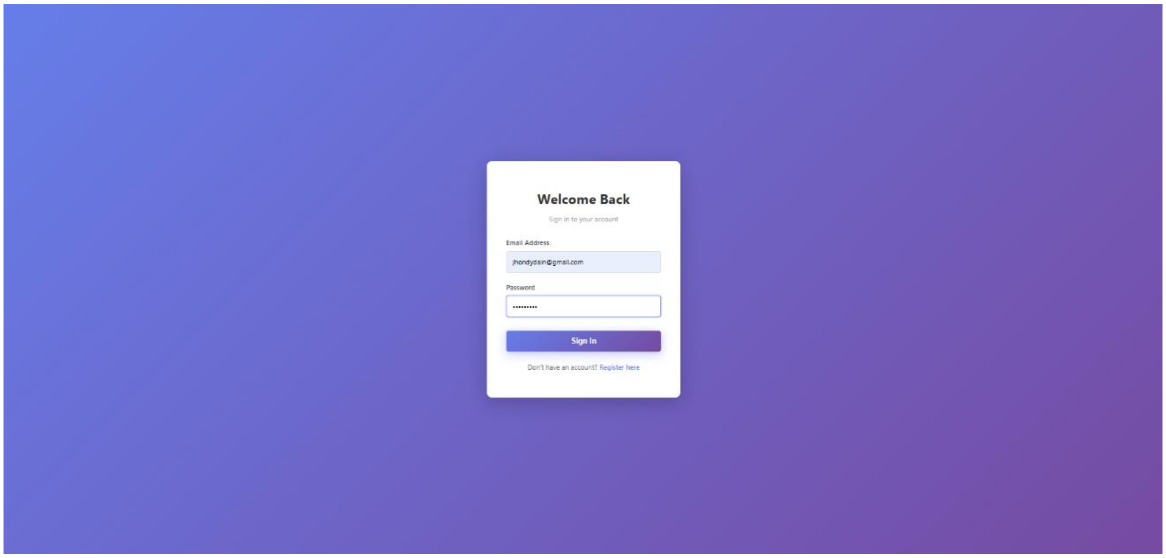
Registration



Dashboard/Profile with logout



Login



6. Appendices